Audio Visualiser DSP Board

https://github.com/drewsum/audio_visualiser

01. Table of Contents

02. 12V Power Input — 12V Power Input — File: POS12_power_input.kicad_sch

03. 12V Telemetry — 12V Telemetry — File: POS12_telemetry.kicad_sch

04. 3.3V Power Supply — 3.3V Power Supply — File: POS3P3_power_supply.kicad_sch

05. 3.3V Telemetry — 3.3V Telemetry — File: POS3P3_telemetry.kicad_sch

06. dsPIC33 Programming — dsPIC33 Programming — File: dspic33_programming.kicad_sch

07. dsPIC33 Power — dsPIC33 Power — File: dspic33_power.kicad_sch

08. dsPIC33 GPIO — dsPIC33 GPIO — File: dspic33_gpio.kicad_sch

09. USB UART Bridge — USB UART Bridge — File: usb_uart_bridge.kicad_sch

10. Platform ETC — Platform ETC — File: platform_etc.kicad_sch

11. PGOOD LEDs — PGOOD LEDs — File: pgood_leds.kicad_sch

12. Status LEDs — Status LEDs — File: status_leds.kicad_sch

13. RCA Signal Conditioning — RCA Signal Conditioning — File: rca_signal_conditioning.kicad_sch

14. Speaker Signal Conditioning — Speaker Signal Conditioning — File: speaker_signal_conditoning.kicad_sch

15. Source Selection, PGAs — Source Selection, PGAs — File: source_selection_pgas.kicad_sch

16. Anti-alias Filtering — Anti-alias Filtering — File: anti_alias_filtering.kicad_sch

17. Audio Voltage References — Audio Voltage References — File: audio_voltage_references.kicad_sch

18. LED Panel Buffers — LED Panel Buffers — File: led_panel_buffers.kicad_sch

19. Pushbuttons — Pushbuttons — File: pushbuttons.kicad_sch

20. Mechanical — Mechanical — File: mechanical.kicad_sch

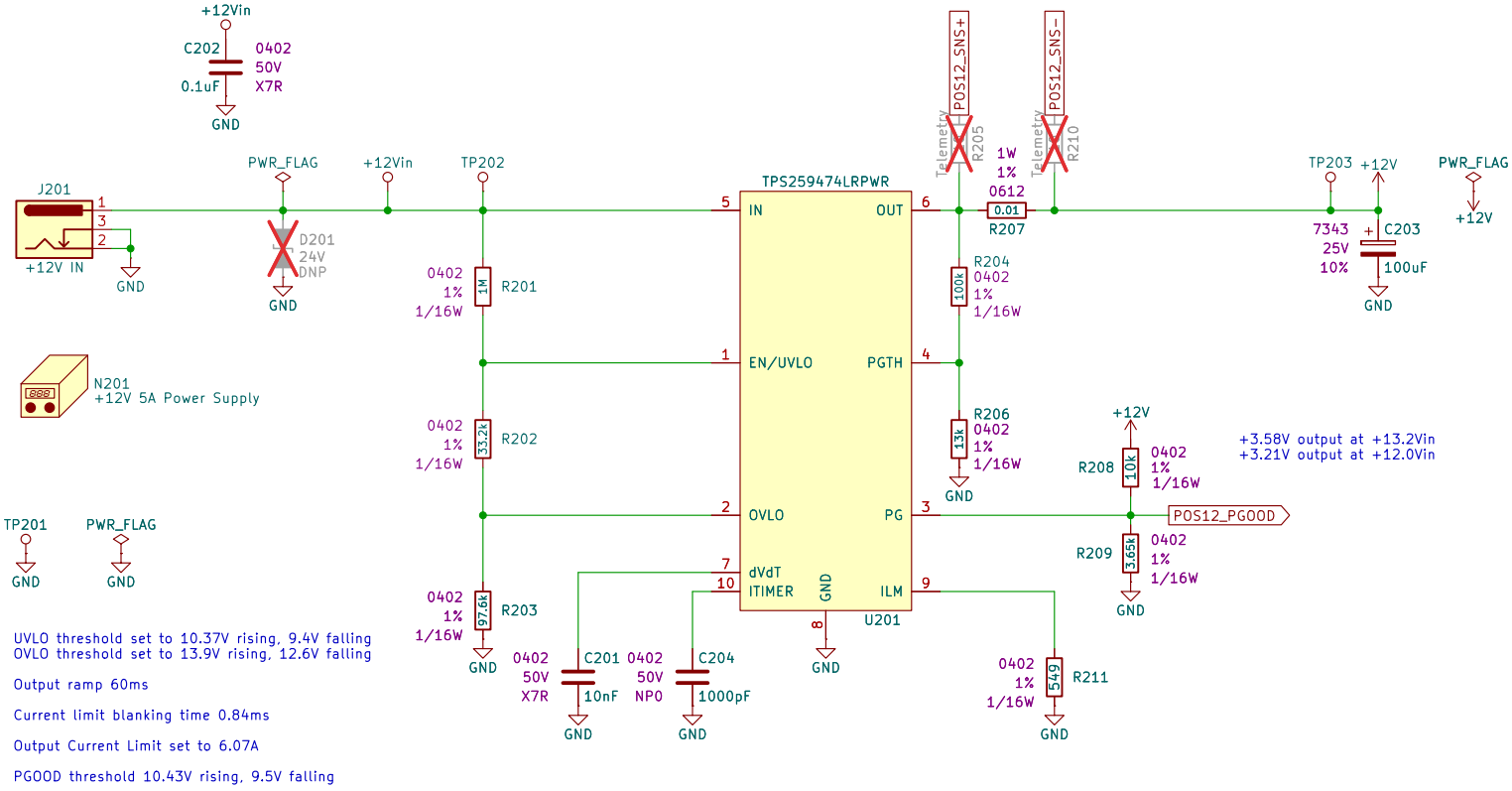
21. LED Panel Connectors — LED Panel Connectors — File: led_panel_connectors.kicad_sch

Drew Maatman and Drew Koetting

Sheet: /
File: Audio_Visualiser_DSP.kicad_sch

Title: Audio Visualiser DSP

Size: A	Date: 2026-10-27	Rev: A
KiCad E.D.A. 10.0.7		Id: 1/21



Load	Draw	Quantity	Total
LED Panel VLED 256 * 0.02A 8 3V = 15.36W / 12V = 1.28A / 80% efficiency	1.6A	2	3.2A
LED Panel POS3P3 1A * 3.3V / 12V / 80% efficiency	0.344A	2	0.688A
Local POS3P3 1A * 3.3V / 12V / 80% efficiency	0.344A	1	0.344A
Total		50.784W	4.232A

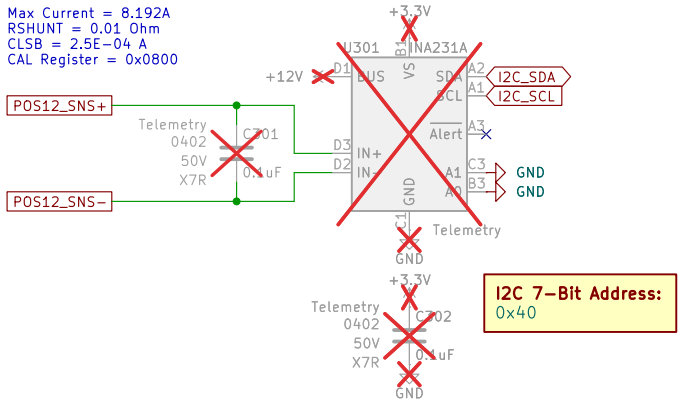
Drew Maatman and Drew Koetting

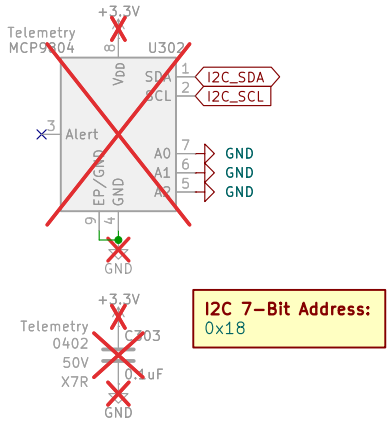
Sheet: /12V Power Input/

File: POS12_power_input.kicad_sch

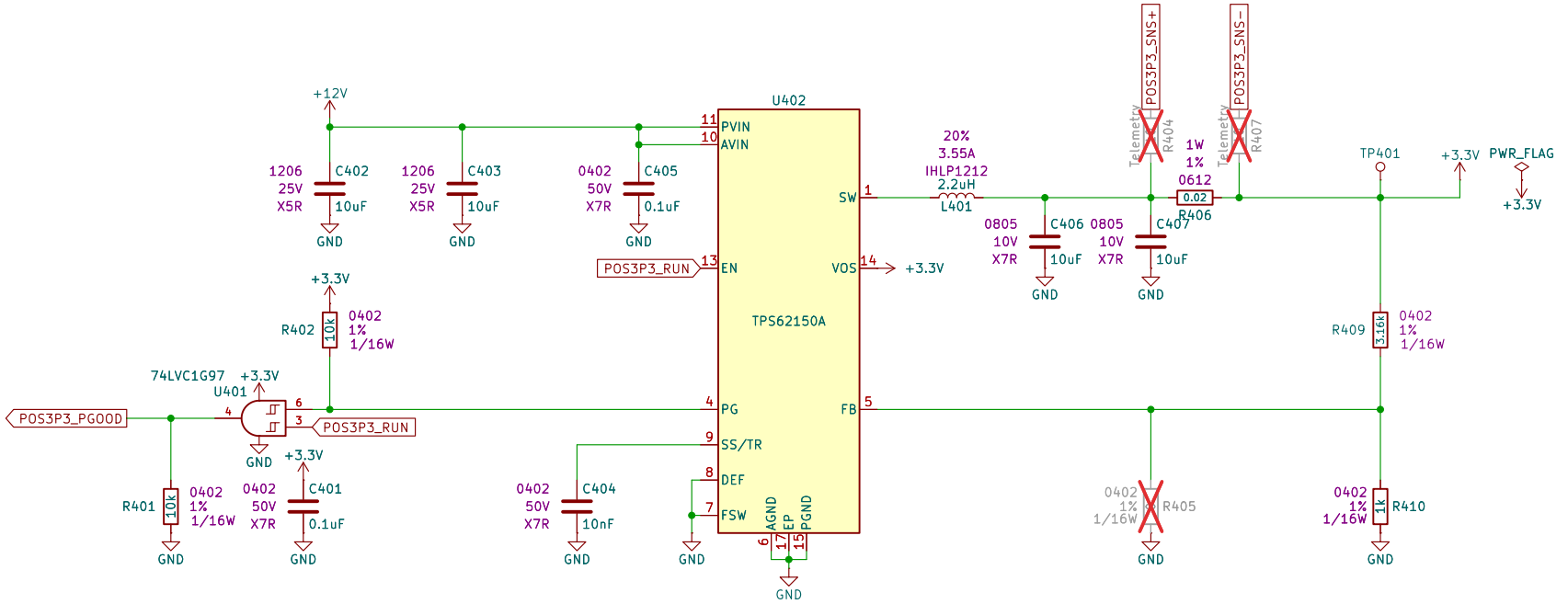
Title: Audio Visualiser DSP

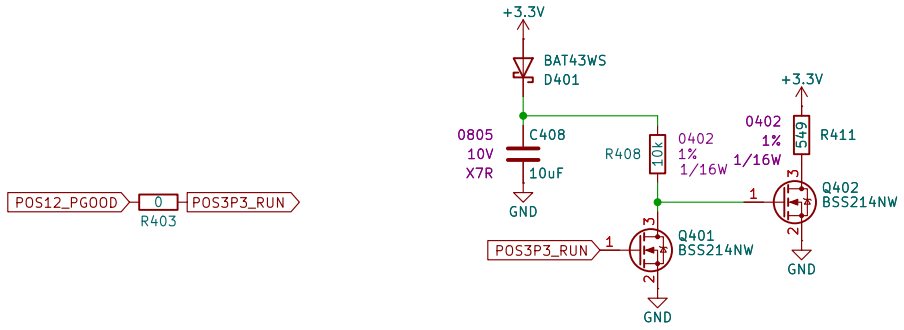
Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 2/21





Drew Maatman and Drew Koetting		
Sheet: /12V Telemetry/		
File: POS12_telemetry.kicad_sch		
Title: Audio Visualiser DSP		
Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 3/21





Load	Draw	Quantity	Total
dsPIC	50mA	1	50mA
8MHz dsPIC Clock	7mA	1	7mA
MCP6N11	1.1mA	6	6.6mA
MCP6024	5.4mA	3	16.2mA
Analog Switch	0.05mA	1	0.05mA
Digipot	0.6mA	1	0.6mA
1.6V Reference	0.2mA	1	0.2mA
PCA9536 GPIO Expander	0.29mA	1	0.29mA
I2C Buffers	8.3mA	3	24.9mA
Octal Buffers	1mA	2	2mA
LVC1G19s	0.01mA	21	0.21mA
Rotary Encoder Module	50mA	1	50mA
LEDs	1.3mA	11	14.3mA
INA231	0.42mA	2	0.84mA
MCP9804	0.4mA	3	1.2mA
ETC	0.36mA	1	0.36mA
Cap Touch Sensor	0.729mA	1	0.729mA
Total		0.579W	175.48mA

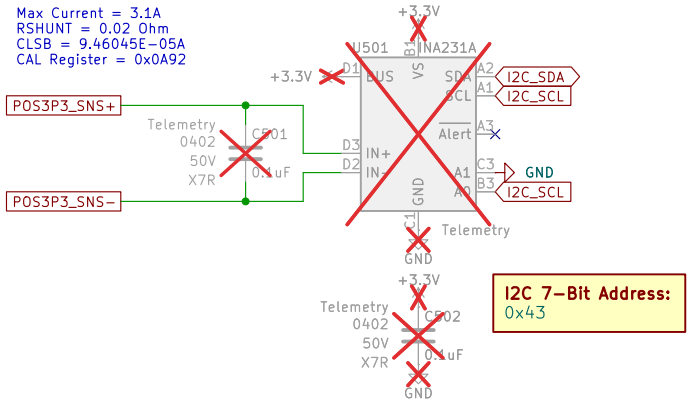
Drew Maatman and Drew Koetting

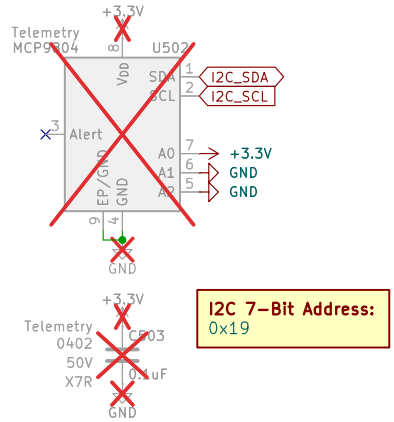
Sheet: /3.3V Power Supply/

File: POS3P3_power_supply.kicad_sch

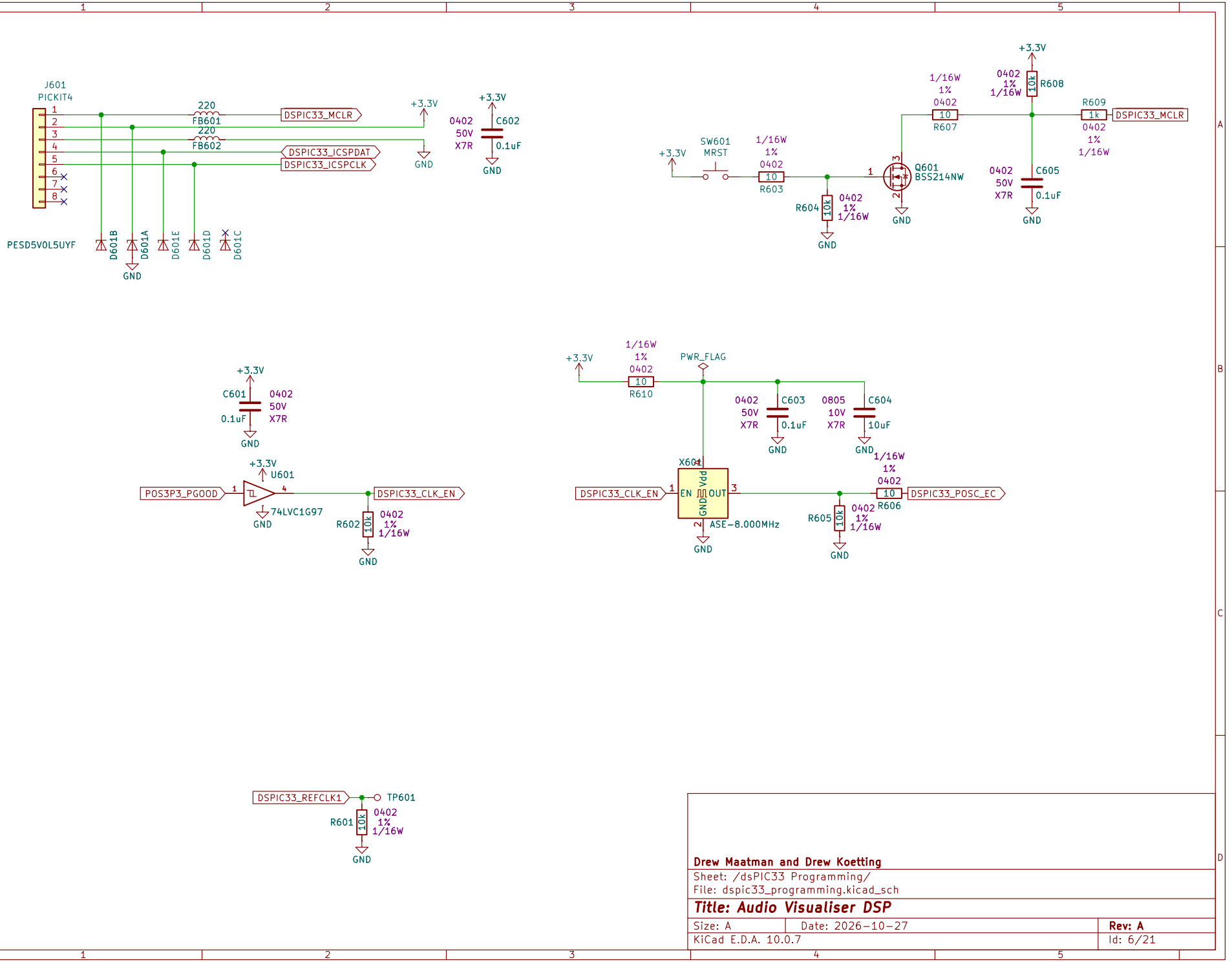
Title: Audio Visualiser DSP

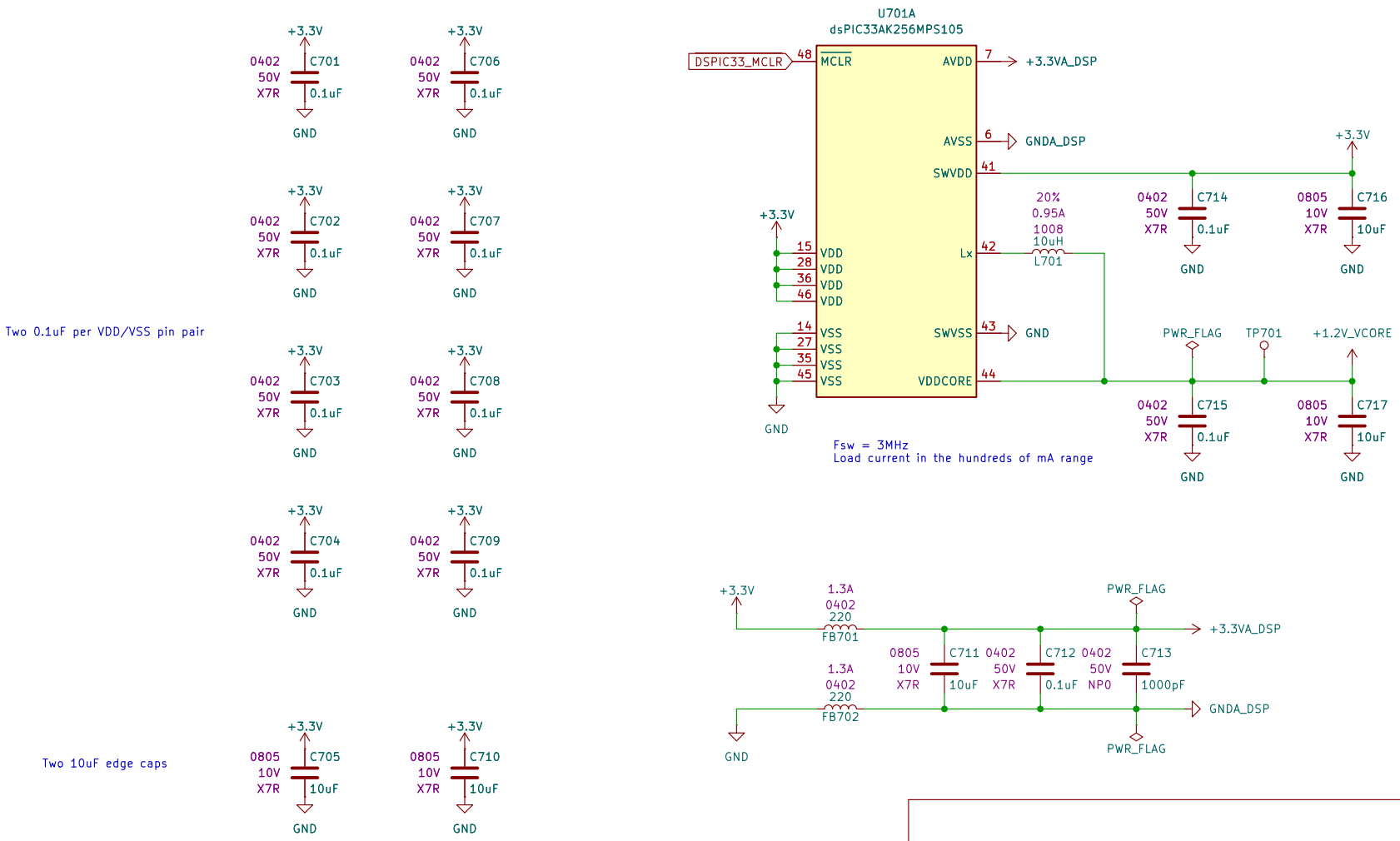
Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 4/21



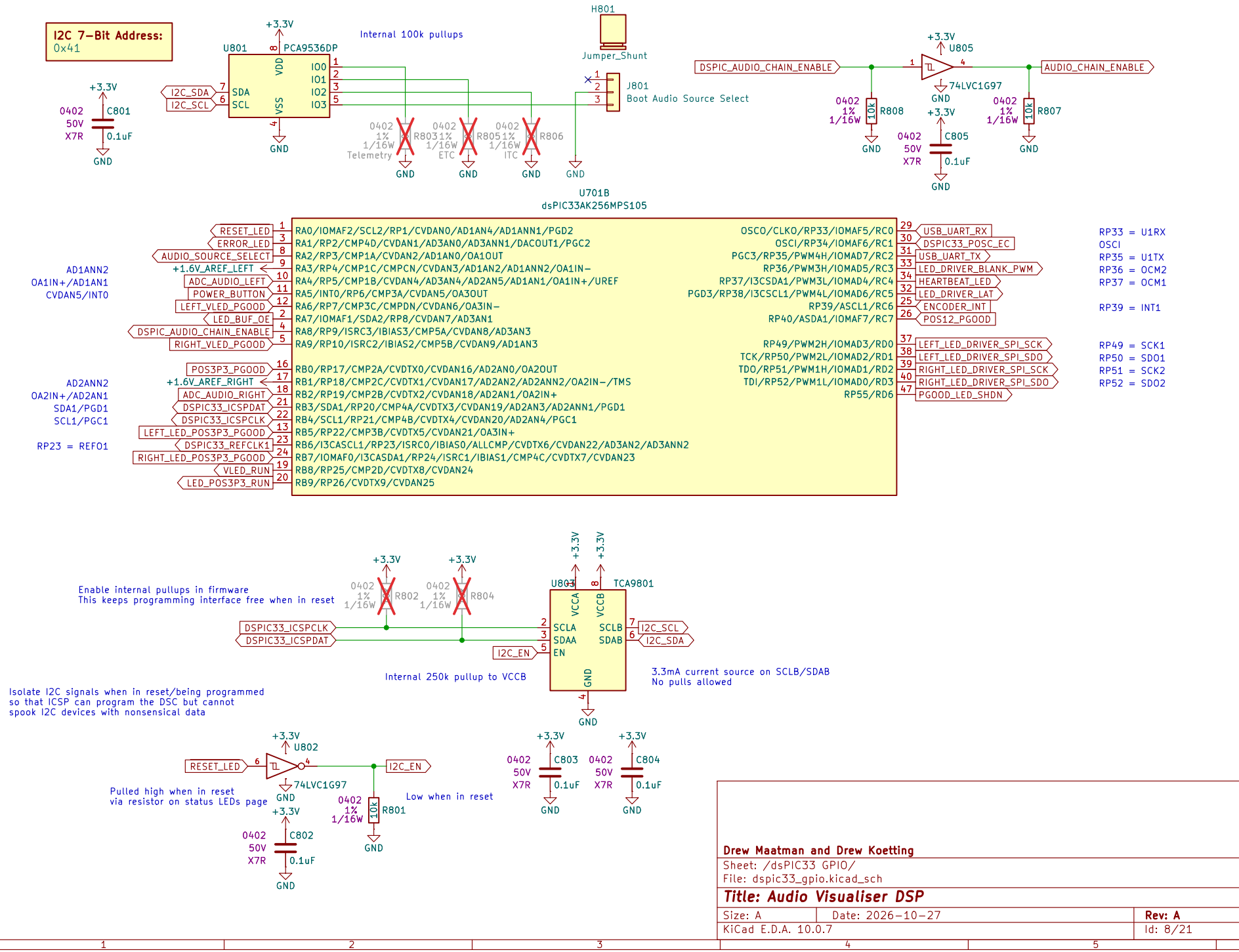


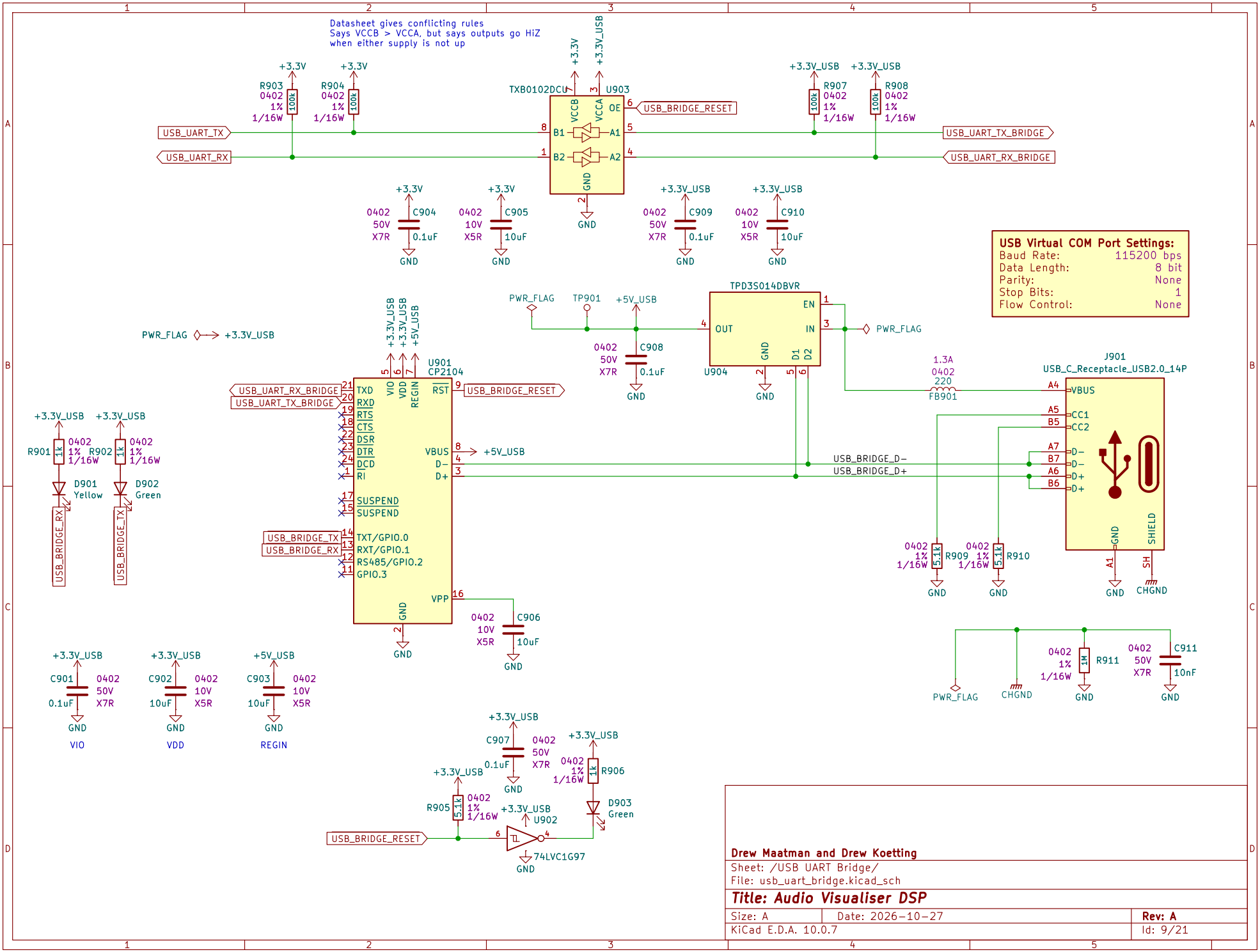
Drew Maatman and Drew Koetting		
Sheet: /3.3V Telemetry/		
File: POS3P3_telemetry.kicad_sch		
Title: Audio Visualiser DSP		
Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 5/21


J601
PICKIT4
1
2
3
4
5
6
7
8
220
FB601
220
FB602
DSPIC33_MCLR
DSPIC33_ICSPDAT
DSPIC33_ICSPCLK
+3.3V
GND
+3.3V
0402
50V
X7R
C602
0.1uF
GND
PESD5V0L5UYF
D601B
D601A
D601E
D601D
D601C
GND
+3.3V
1/16W
1%
0402
10
R607
0402
1%
1/16W
10k
R608
R609
1k
DSPIC33_MCLR
0402
1%
1/16W
+3.3V
SW601
MRST
1/16W
1%
0402
10
R603
Q601
BSS214NW
R604
10k
0402
1%
1/16W
GND
0402
50V
X7R
C605
0.1uF
GND
+3.3V
C601
0402
50V
X7R
0.1uF
GND
+3.3V
U601
POS3P3_PGOOD
1
TL
4
DSPIC33_CLK_EN
74LVC1G97
GND
R602
10k
0402
1%
1/16W
GND
+3.3V
1/16W
1%
0402
10
R610
PWR_FLAG
0402
50V
X7R
C603
0.1uF
GND
0805
10V
X7R
C604
10uF
GND
X601
DSPIC33_CLK_EN
1
EN
Vdd
OUT
3
GND
2
ASE-8.000MHz
GND
1/16W
1%
0402
10
R606
DSPIC33_POSC_EC
R605
10k
0402
1%
1/16W
GND
DSPIC33_REFCLK1
TP601
R601
10k
0402
1%
1/16W
GND
Drew Maatman and Drew Koetting
Sheet: /dsPIC33 Programming/
File: dspic33_programming.kicad_sch
Title: Audio Visualiser DSP
Size: A
Date: 2026-10-27
Rev: A
KiCad E.D.A. 10.0.7
Id: 6/21

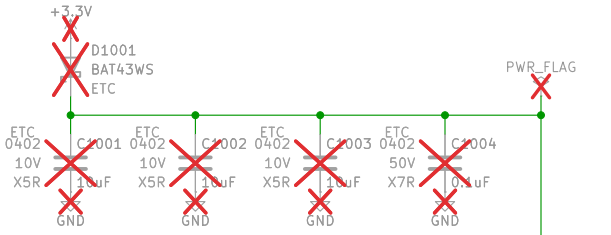


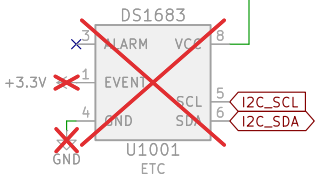
Drew Maatman and Drew Koetting		
Sheet: /dsPIC33 Power/		
File: dspic33_power.kicad_sch		
Title: Audio Visualiser DSP		
Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 7/21


I2C 7-Bit Address:
0x41
U801
PCA9536DP
+3.3V
Internal 100k pullups
I2C_SDA
I2C_SCL
SDA
SCL
VDD
VSS
IO0
IO1
IO2
IO3
GND
C801
0402
50V
X7R
0.1uF
H801
Jumper_Shunt
J801
Boot Audio Source Select
R803
1%
1/16W
Telemetry
R805
ETC
R806
ITC
U701B
dsPIC33AK256MPS105
DSPIC_AUDIO_CHAIN_ENABLE
U805
74LVC1G97
AUDIO_CHAIN_ENABLE
R808
R807
C805
RESET_LED
ERROR_LED
AUDIO_SOURCE_SELECT
+1.6V_AREF_LEFT
ADC_AUDIO_LEFT
POWER_BUTTON
LEFT_VLED_PGOOD
LED_BUF_OE
DSPIC_AUDIO_CHAIN_ENABLE
RIGHT_VLED_PGOOD
POS3P3_PGOOD
+1.6V_AREF_RIGHT
ADC_AUDIO_RIGHT
DSPIC33_ICSPDAT
DSPIC33_ICSPCLK
LEFT_LED_POS3P3_PGOOD
DSPIC33_REFCLK1
RIGHT_LED_POS3P3_PGOOD
VLED_RUN
LED_POS3P3_RUN
RA0/IOMAF2/SCL2/RP1/CVDAN0/AD1AN4/AD1ANN1/PGD2
RA1/RP2/CMP4D/CVDAN1/AD3AN0/AD3ANN1/DACOUT1/PGC2
RA2/RP3/CMP1A/CVDAN2/AD1AN0/OA1OUT
RA3/RP4/CMP1C/CMPCN/CVDAN3/AD1AN2/AD1ANN2/OA1IN−
RA4/RP5/CMP1B/CVDAN4/AD3AN4/AD2AN5/AD1AN1/OA1IN+/UREF
RA5/INT0/RP6/CMP3A/CVDAN5/OA3OUT
RA6/RP7/CMP3C/CMPDN/CVDAN6/OA3IN−
RA7/IOMAF1/SDA2/RP8/CVDAN7/AD3AN1
RA8/RP9/ISRC3/IBIAS3/CMP5A/CVDAN8/AD3AN3
RA9/RP10/ISRC2/IBIAS2/CMP5B/CVDAN9/AD1AN3
RB0/RP17/CMP2A/CVDTX0/CVDAN16/AD2AN0/OA2OUT
RB1/RP18/CMP2C/CVDTX1/CVDAN17/AD2AN2/AD2ANN2/OA2IN−/TMS
RB2/RP19/CMP2B/CVDTX2/CVDAN18/AD2AN1/OA2IN+
RB3/SDA1/RP20/CMP4A/CVDTX3/CVDAN19/AD2AN3/AD2ANN1/PGD1
RB4/SCL1/RP21/CMP4B/CVDTX4/CVDAN20/AD2AN4/PGC1
RB5/RP22/CMP3B/CVDTX5/CVDAN21/OA3IN+
RB6/I3CASCL1/RP23/ISRC0/IBIAS0/ALLCMP/CVDTX6/CVDAN22/AD3AN2/AD3ANN2
RB7/IOMAF0/I3CASDA1/RP24/ISRC1/IBIAS1/CMP4C/CVDTX7/CVDAN23
RB8/RP25/CMP2D/CVDTX8/CVDAN24
RB9/RP26/CVDTX9/CVDAN25
OSCO/CLKO/RP33/IOMAF5/RC0
OSCI/RP34/IOMAF6/RC1
PGC3/RP35/PWM4H/IOMAD7/RC2
RP36/PWM3H/IOMAD5/RC3
RP37/I3CSDA1/PWM3L/IOMAD4/RC4
PGD3/RP38/I3CSCL1/PWM4L/IOMAD6/RC5
RP39/ASCL1/RC6
RP40/ASDA1/IOMAF7/RC7
RP49/PWM2H/IOMAD3/RD0
TCK/RP50/PWM2L/IOMAD2/RD1
TDO/RP51/PWM1H/IOMAD1/RD2
TDI/RP52/PWM1L/IOMAD0/RD3
RP55/RD6
USB_UART_RX
DSPIC33_POSC_EC
USB_UART_TX
LED_DRIVER_BLANK_PWM
HEARTBEAT_LED
LED_DRIVER_LAT
ENCODER_INT
POS12_PGOOD
LEFT_LED_DRIVER_SPI_SCK
LEFT_LED_DRIVER_SPI_SDO
RIGHT_LED_DRIVER_SPI_SCK
RIGHT_LED_DRIVER_SPI_SDO
PGOOD_LED_SHDN
AD1ANN2
OA1IN+/AD1AN1
CVDAN5/INT0
AD2ANN2
OA2IN+/AD2AN1
SDA1/PGD1
SCL1/PGC1
RP23 = REFO1
RP33 = U1RX
OSCI
RP35 = U1TX
RP36 = OCM2
RP37 = OCM1
RP39 = INT1
RP49 = SCK1
RP50 = SDO1
RP51 = SCK2
RP52 = SDO2
Enable internal pullups in firmware
This keeps programming interface free when in reset
R802
R804
U803
TCA9801
VCCA
VCCB
SCLA
SDAA
EN
SCLB
SDAB
GND
I2C_EN
Internal 250k pullup to VCCB
3.3mA current source on SCLB/SDAB
No pulls allowed
Isolate I2C signals when in reset/being programmed
so that ICSP can program the DSC but cannot
spook I2C devices with nonsensical data
U802
Pulled high when in reset
via resistor on status LEDs page
Low when in reset
R801
C802
C803
C804
Drew Maatman and Drew Koetting
Sheet: /dsPIC33 GPIO/
File: dspic33_gpio.kicad_sch
Title: Audio Visualiser DSP
Size: A
Date: 2026-10-27
Rev: A
KiCad E.D.A. 10.0.7
Id: 8/21

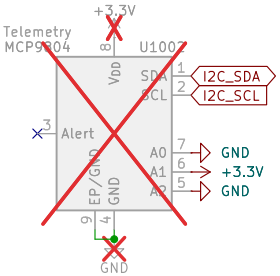


Drew Maatman and Drew Koetting		
Sheet: /USB UART Bridge/		
File: usb_uart_bridge.kicad_sch		
Title: Audio Visualiser DSP		
Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 9/21





I2C 7-Bit Address:
0x6B



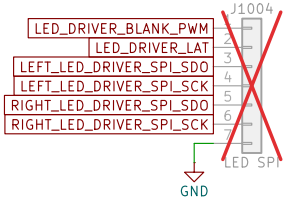


I2C 7-Bit Address:
0x1A

Ambient, place away from power components

Test Points



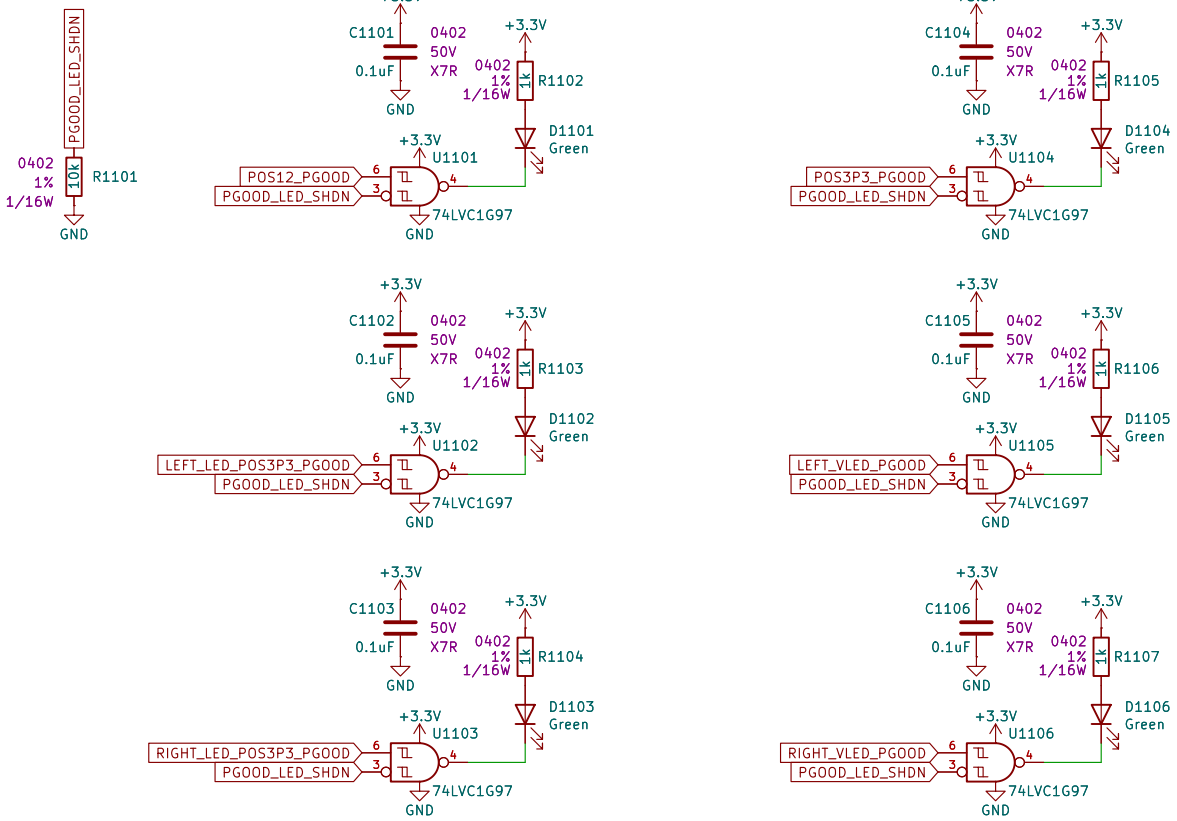


Drew Maatman and Drew Koetting

Sheet: /Platform ETC/
File: platform_etc.kicad_sch

Title: Audio Visualiser DSP

Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 10/21

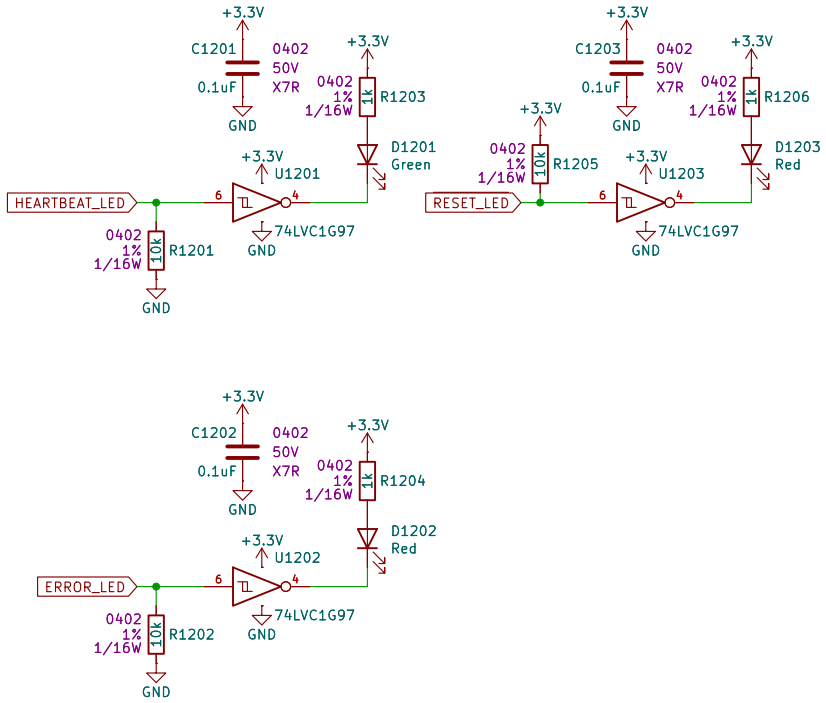


Drew Maatman and Drew Koetting

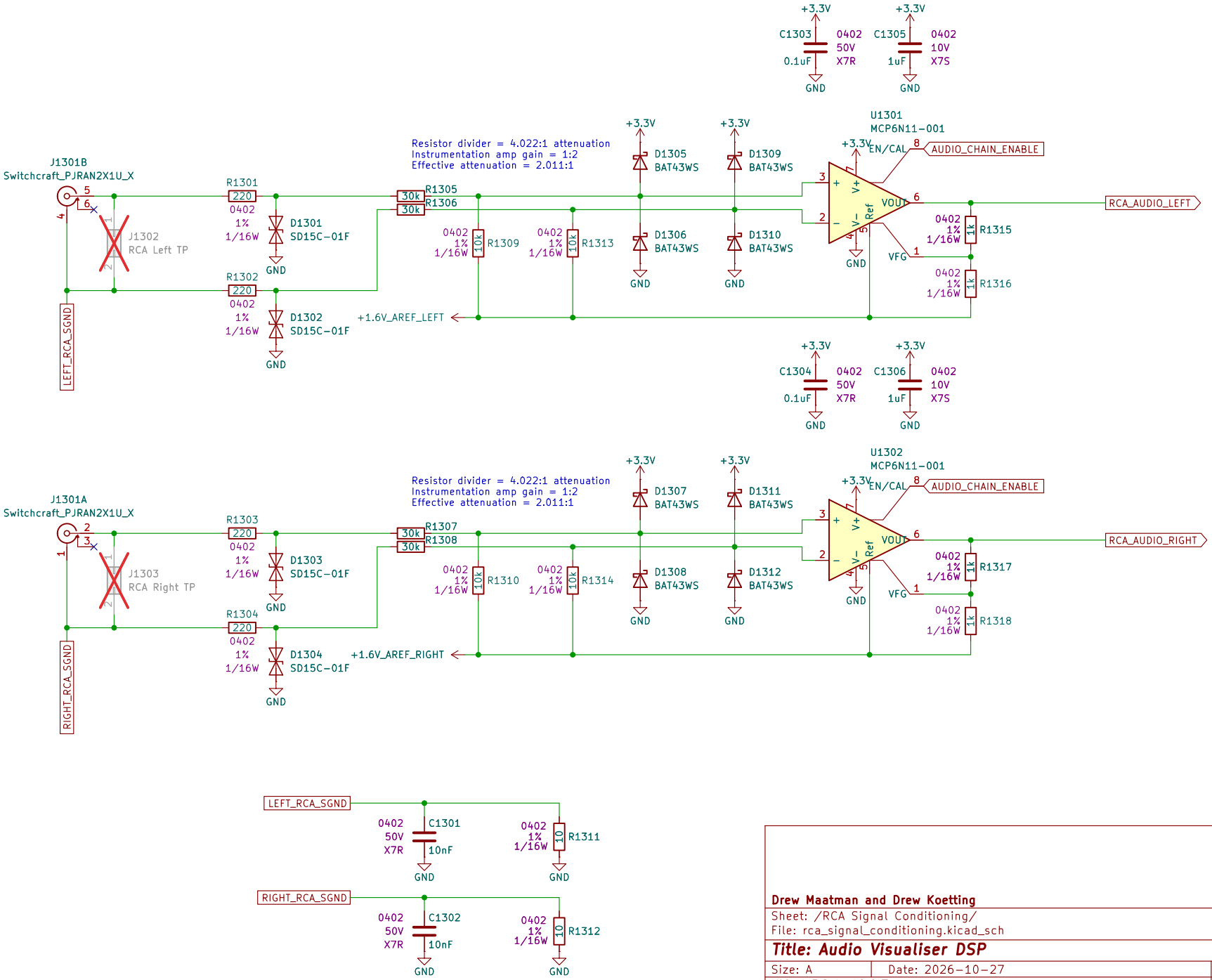
Sheet: /PGOOD LEDs/
File: pgood_leds.kicad_sch

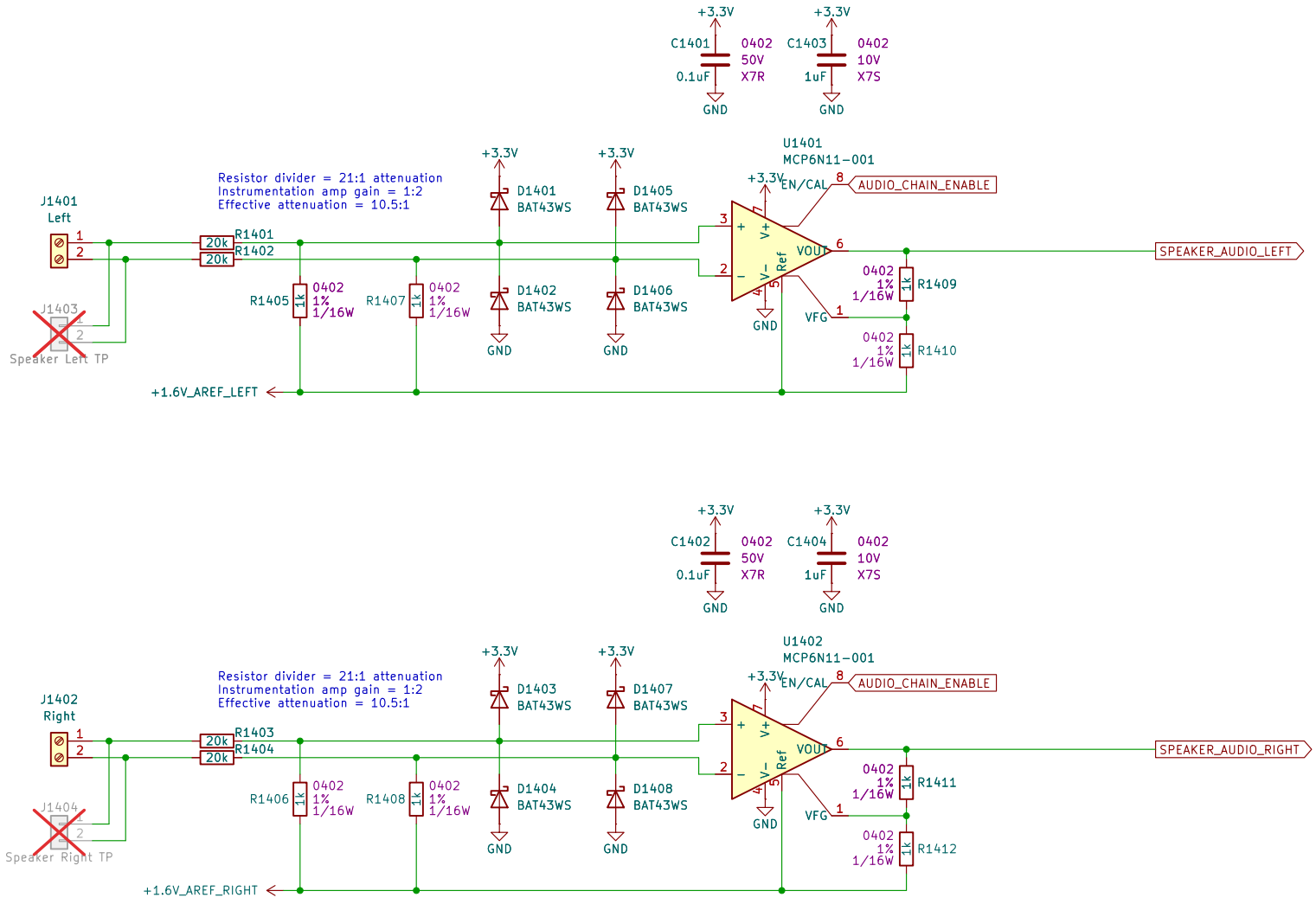
Title: Audio Visualiser DSP

Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 11/21

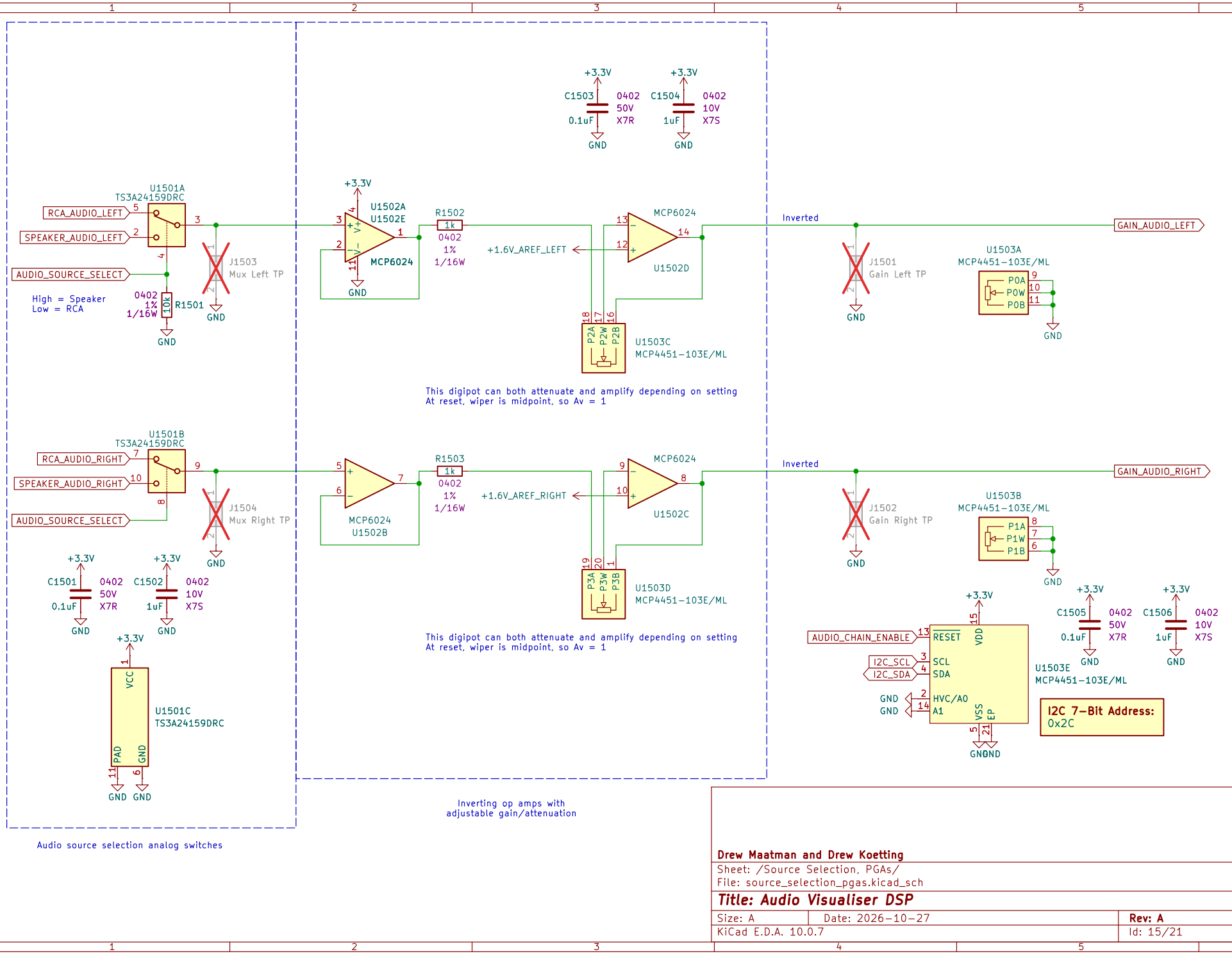


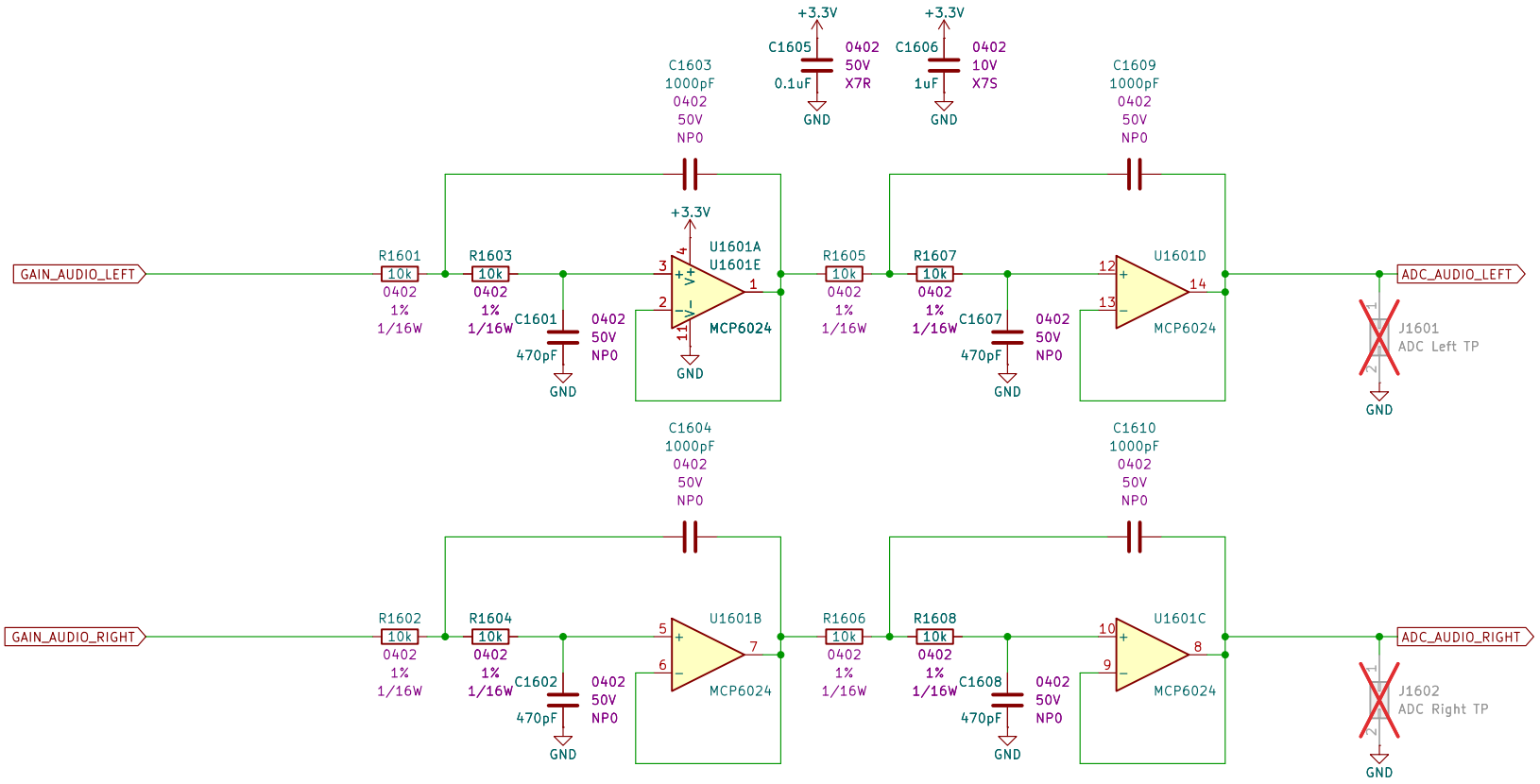
Drew Maatman and Drew Koetting		
Sheet: /Status LEDs/		
File: status_leds.kicad_sch		
Title: Audio Visualiser DSP		
Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 12/21


+3.3V
C1303
0402
50V
X7R
0.1uF
GND
+3.3V
C1305
0402
10V
X7S
1uF
GND
U1301
MCP6N11-001
EN/CAL
AUDIO_CHAIN_ENABLE
VOUT
VFG
V+
V-
Ref
RCA_AUDIO_LEFT
J1301B
Switchcraft_PJRAN2X1U_X
J1302
RCA Left TP
LEFT_RCA_SGND
R1301
220
0402
1%
1/16W
R1302
D1301
SD15C-01F
D1302
Resistor divider = 4.022:1 attenuation
Instrumentation amp gain = 1:2
Effective attenuation = 2.011:1
R1305
30k
R1306
R1309
10k
R1313
D1305
BAT43WS
D1309
D1306
D1310
R1315
1k
R1316
+1.6V_AREF_LEFT
C1304
C1306
U1302
MCP6N11-001
RCA_AUDIO_RIGHT
J1301A
Switchcraft_PJRAN2X1U_X
J1303
RCA Right TP
RIGHT_RCA_SGND
R1303
R1304
D1303
D1304
R1307
R1308
R1310
R1314
D1307
D1311
D1308
D1312
R1317
R1318
+1.6V_AREF_RIGHT
LEFT_RCA_SGND
C1301
10nF
R1311
10
RIGHT_RCA_SGND
C1302
R1312
Drew Maatman and Drew Koetting
Sheet: /RCA Signal Conditioning/
File: rca_signal_conditioning.kicad_sch
Title: Audio Visualiser DSP
Size: A
Date: 2026-10-27
Rev: A
KiCad E.D.A. 10.0.7
Id: 13/21



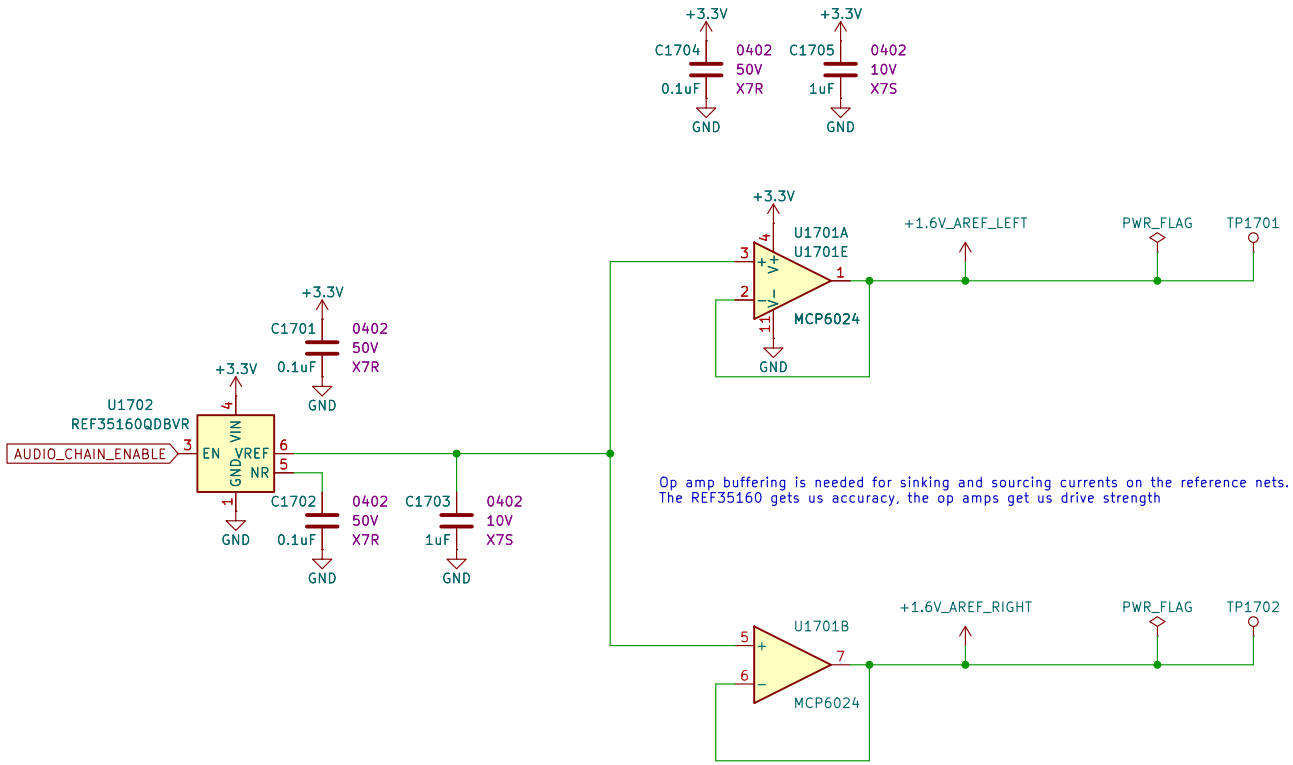
Drew Maatman and Drew Koetting		
Sheet: /Speaker Signal Conditioning/		
File: speaker_signal_conditoning.kicad_sch		
Title: Audio Visualiser DSP		
Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 14/21


RCA_AUDIO_LEFT
SPEAKER_AUDIO_LEFT
AUDIO_SOURCE_SELECT
High = Speaker
Low = RCA
U1501A
TS3A24159DRC
R1501
10k
0402
1%
1/16W
J1503
Mux Left TP
U1502A
U1502E
MCP6024
R1502
1k
+1.6V_AREF_LEFT
U1502D
U1503C
MCP4451-103E/ML
This digipot can both attenuate and amplify depending on setting
At reset, wiper is midpoint, so Av = 1
Inverted
GAIN_AUDIO_LEFT
J1501
Gain Left TP
U1503A
MCP4451-103E/ML
RCA_AUDIO_RIGHT
SPEAKER_AUDIO_RIGHT
AUDIO_SOURCE_SELECT
U1501B
TS3A24159DRC
J1504
Mux Right TP
U1502B
R1503
+1.6V_AREF_RIGHT
U1502C
U1503D
GAIN_AUDIO_RIGHT
J1502
Gain Right TP
U1503B
C1501
C1502
C1503
C1504
C1505
C1506
0.1uF
1uF
50V
10V
X7R
X7S
U1501C
TS3A24159DRC
U1503E
MCP4451-103E/ML
AUDIO_CHAIN_ENABLE
I2C_SCL
I2C_SDA
I2C 7-Bit Address:
0x2C
Audio source selection analog switches
Inverting op amps with
adjustable gain/attenuation
Drew Maatman and Drew Koetting
Sheet: /Source Selection, PGAs/
File: source_selection_pgas.kicad_sch
Title: Audio Visualiser DSP
Size: A
Date: 2026-10-27
Rev: A
KiCad E.D.A. 10.0.7
Id: 15/21

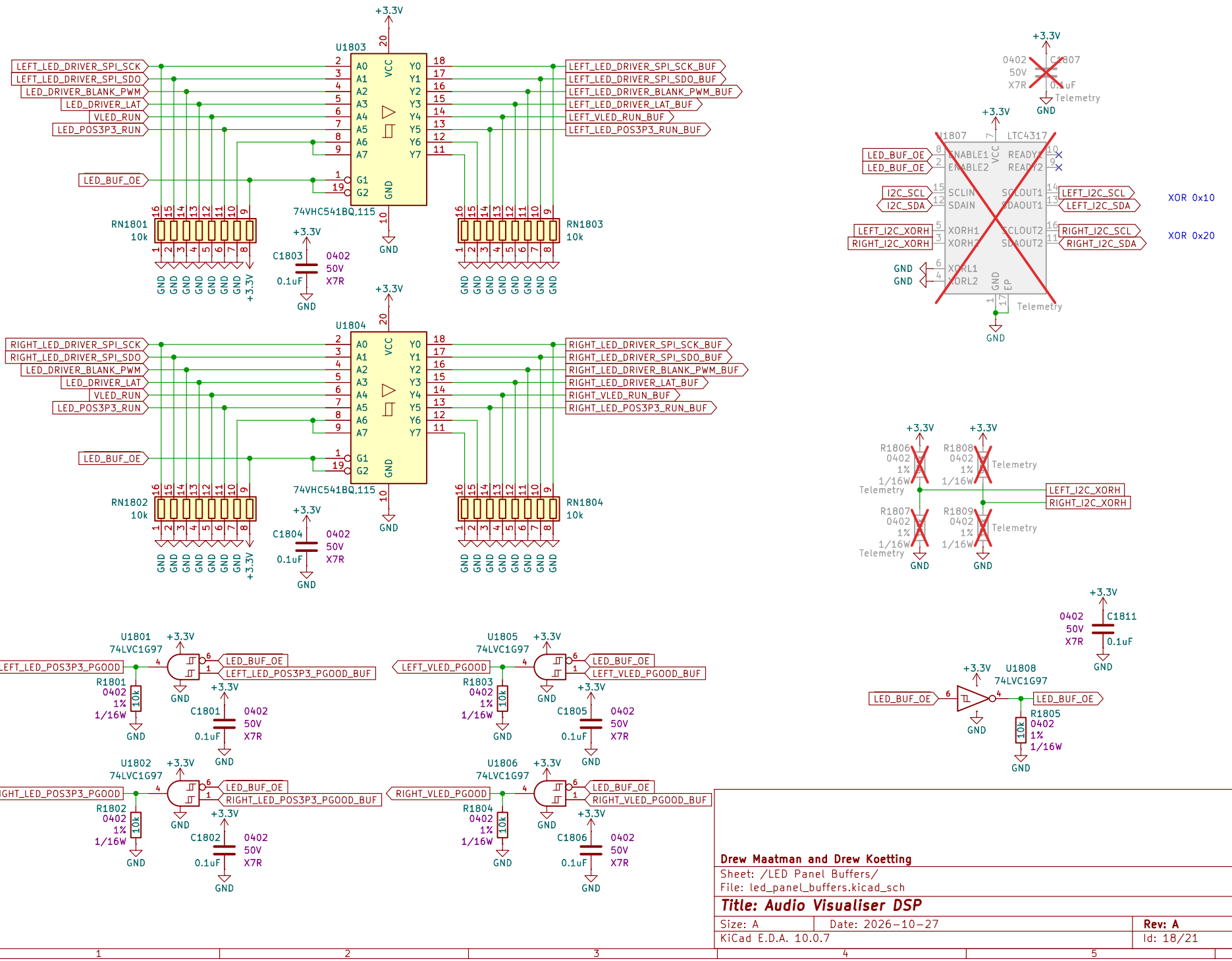


4-pole Sallen-Key active anti-aliasing lowpass filters
Unity gain
fc = 23.2kHz, Q = 0.707 each
If sampling at 44.1kHz, nyquist is 22.05kHz
Must filter hard around this frequency

Drew Maatman and Drew Koetting		
Sheet: /Anti-alias Filtering/		
File: anti_alias_filtering.kicad_sch		
Title: Audio Visualiser DSP		
Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 16/21

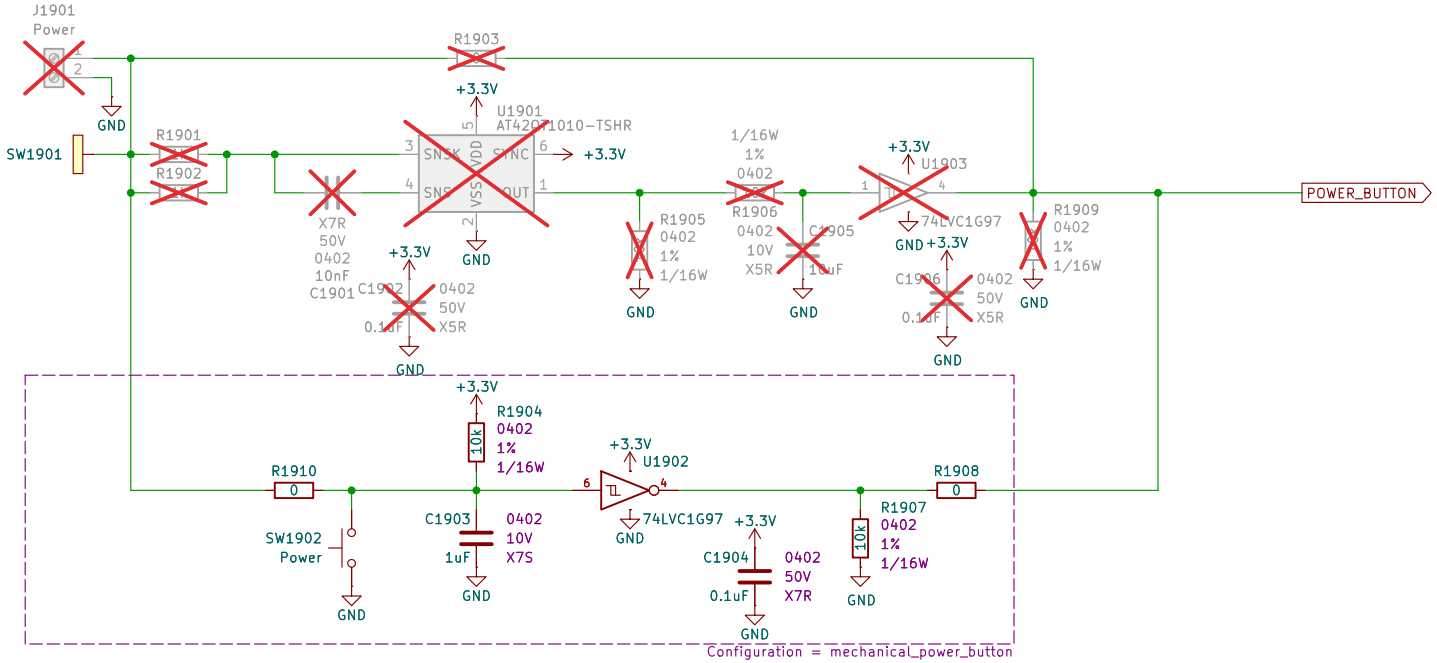


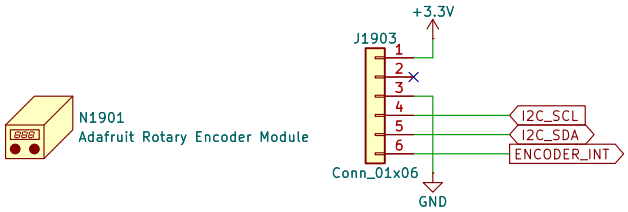
Drew Maatman and Drew Koetting		
Sheet: /Audio Voltage References/		
File: audio_voltage_references.kicad_sch		
Title: Audio Visualiser DSP		
Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 17/21



Drew Maatman and Drew Koetting	
Sheet: /LED Panel Buffers/	
File: led_panel_buffers.kicad_sch	
Title: Audio Visualiser DSP	
Size: A	Date: 2026-10-27
KiCad E.D.A. 10.0.7	Id: 18/21

Rev: A





I2C 7-Bit Address:
0x36

Drew Maatman and Drew Koetting

Sheet: /Pushbuttons/
File: pushbuttons.kicad_sch

Title: Audio Visualiser DSP

Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 19/21

H2001
MountingHole

H2003
MountingHole

H2002
MountingHole

H2004
MountingHole

Screw
MK2001

Screw
MK2005

Screw
MK2002

Screw
MK2006

Standoff
MK2003

Standoff
MK2007

Standoff
MK2004

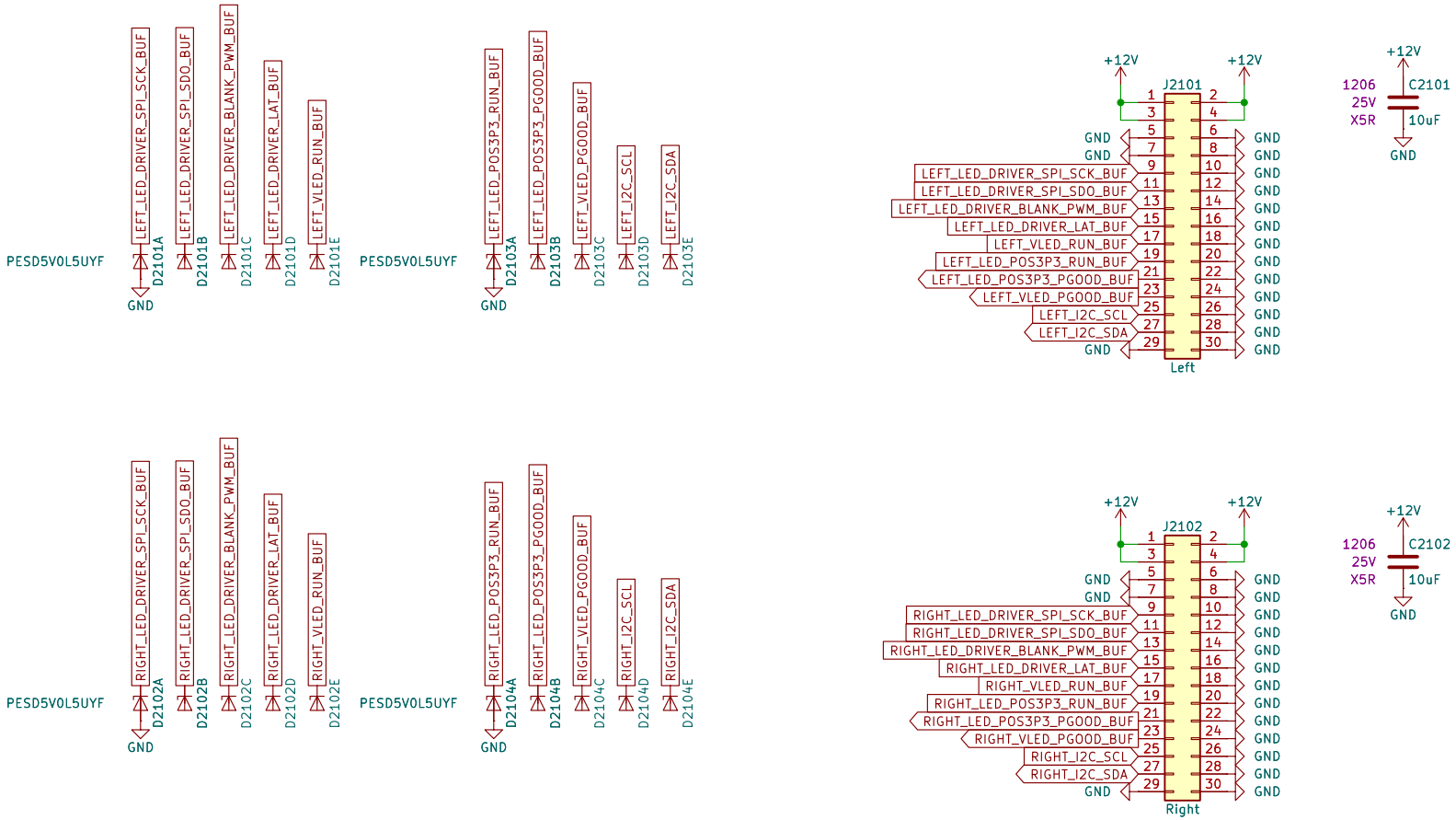
Standoff
MK2008

Drew Maatman and Drew Koetting

Sheet: /Mechanical/
File: mechanical.kicad_sch

Title: Audio Visualiser DSP

Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 20/21



Drew Maatman and Drew Koetting

Sheet: /LED Panel Connectors/
File: led_panel_connectors.kicad_sch

Title: Audio Visualiser DSP

Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 21/21